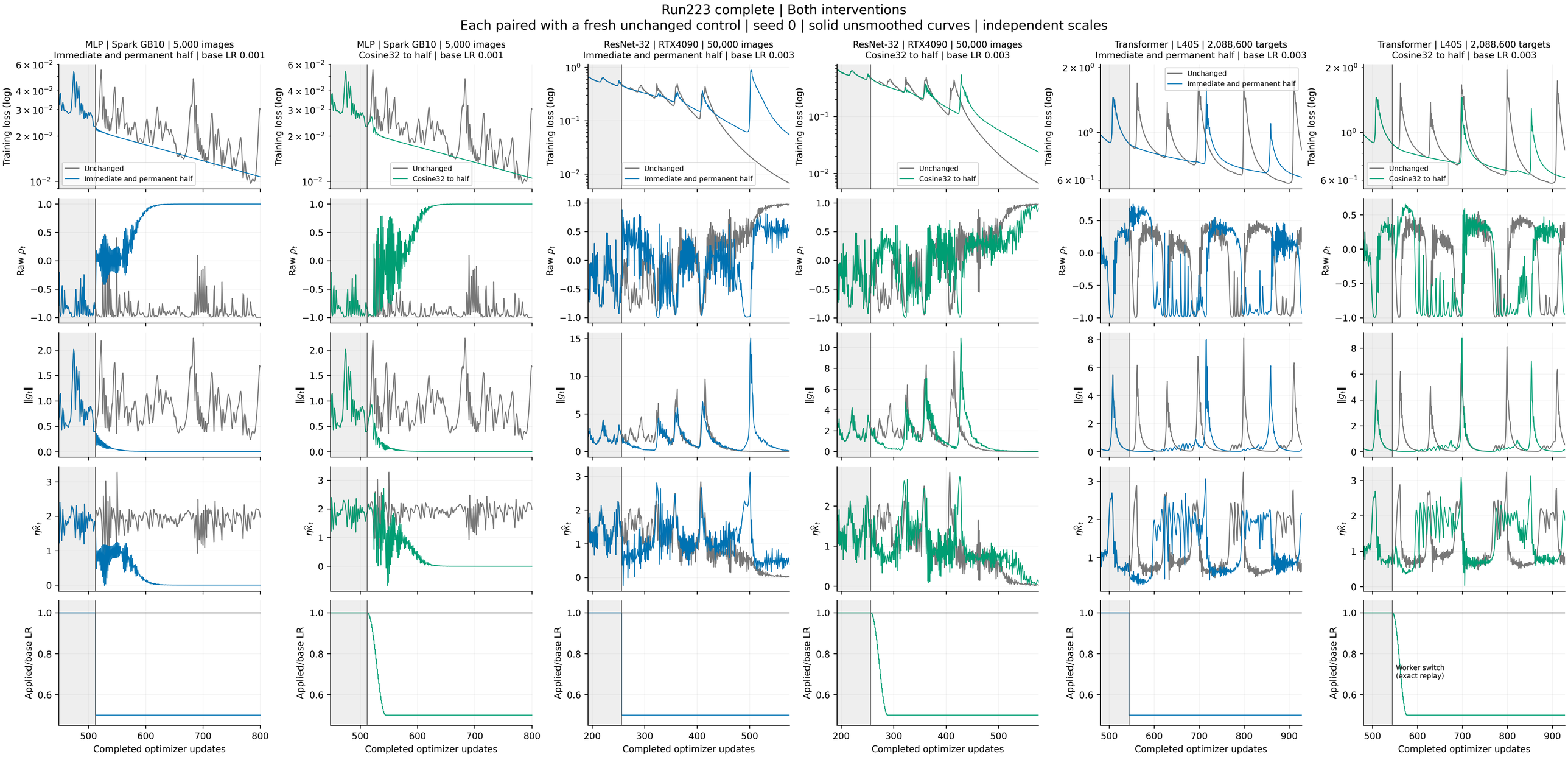

Run223 complete | Both interventions
Each paired with a fresh unchanged control | seed 0 | solid unsmoothed curves | independent scales
MLP | Spark GB10 | 5,000 images
Immediate and permanent half | base LR 0.001
MLP | Spark GB10 | 5,000 images
Cosine32 to half | base LR 0.001
ResNet-32 | RTX4090 | 50,000 images
Immediate and permanent half | base LR 0.003
ResNet-32 | RTX4090 | 50,000 images
Cosine32 to half | base LR 0.003
Transformer | L40S | 2,088,600 targets
Immediate and permanent half | base LR 0.003
Transformer | L40S | 2,088,600 targets
Cosine32 to half | base LR 0.003
Unchanged
Immediate and permanent half
Cosine32 to half
Training loss (log)
Raw ρ_t
$\|g_t\|$
$\eta\hat{\kappa}_t$
Applied/base LR
Worker switch
(exact replay)
Completed optimizer updates
Gray region: shared prefix; vertical line: intervention fork. Transformer cosine switches L40S workers at544 after exact replay.
Other curves stay on one worker. Historical control reuse failed in all three cases; no historical bitwise replay is claimed.